

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Seventh Semester of B. Tech. Examination (EC)****March-April 2018****EC 402/EC 402.01 RF & Microwave Engineering****Date: 28.03.2018, Wednesday****Time: 10.00a.m. To 1.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION I**Q-1** Answer the following questions.

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|----------------|---|---|
| (a) | A coaxial RF cable has a characteristic impedance of 75Ω and C equal to 10 pF/m . The inductance is_____. | 1 |
| (b) | When ZL is connected to lossless line of $Z_0=50 \Omega$, it results in VSWR of 2. The ZL is _____. | 1 |
| (c) | X band frequency range is _____. | 1 |
| (d) | If the short circuited and open circuit impedance of line are 5 and 20Ω respectively the characteristics impedance is given by _____. | 1 |
| (e) | The loss less quarter wave length long, RF transmission of $Z_0 50\Omega$ is terminated in pure resistive load impedance of 100Ω , the input impedance is_____. | 1 |
| (f) | Why TEM mode can't propagate in wave guide? | 2 |
| (g) | A loss less line of characteristic impedance Z_0 is terminated in pure reactance of $-jZ_0$ value. VSWR is _____. | 1 |
| (h) | If a lossless line is terminated with a load impedance of $40+j30 \Omega$, then the Z_0 of the line for minimum value of VSWR will be_____. | 1 |
| (i) | A lossless line has $Z_0= 100 \Omega$, inductance per unit length is $1\mu\text{H/m}$. If the line is operated at 1GHz , the propagation constant is_____. | 1 |
| (j) | When a lossless line of length $\frac{\lambda}{8}$ is short circuited at one end, the input impedance seen at the other end is $j50 \Omega$. If short circuit is now replaced by a load resistance of 75Ω , the magnitude of reflection coefficient is_____. | 1 |
| Q-2 (a) | Explain and draw different types of planar transmission line structures. | 6 |
| (b) | Draw appropriate diagram of 2 cavity klystron amplifier and explain amplification process along with apple gate diagram. | 6 |

OR

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|----------------|---|---|
| Q-2 (a) | Explain E plane, H plane Tee and its S matrix. | 6 |
| (b) | Explain circulator and its S matrix. | 6 |
| Q-3 (a) | Explain reflex klystron and its characteristic modes. | 6 |
| (b) | A microstrip line has following parameters $\epsilon_r=4.4$, $h=1.6\text{mm}$, $t=0.07112\text{mm}$, $w=3.05\text{mm}$, calculate Z_0 . | 2 |
| (c) | A transmission line has characteristic impedance of 50Ω and is terminated in a load impedance of $25+j25 \Omega$. Calculate reflection coefficient and standing wave ration. | 4 |

OR

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|----------------|---|---|
| Q-3 (a) | Explain schottky diode. | 6 |
| (b) | Explain PIN diode. Draw its IV characteristics and explain in detail. | 6 |

SECTION II

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|-------------|-----|--|----|
| Q-4 | (a) | What is the primary use of the TWT? | 1 |
| | (b) | Explain in detail electron bunching process with help of diagram. | 2 |
| | (c) | Why strapping is used in magnetron? | 1 |
| | (d) | What structure in the TWT delays the forward progress of the traveling wave? | 1 |
| | (e) | Draw space charge wheel effect in magnetron. | 1 |
| | (f) | Draw the equivalent circuit of hole and slot cavity of magnetron. | 1 |
| | (g) | E plane and H plane tee junction are also called as _____ and _____ tee junction. | 1 |
| | (h) | Which diode is used in voltage controlled oscillator? | 1 |
| | (i) | Explain standing waves in transmission line. | 2 |
| Q -5 | (a) | Derive the wave equation with complete solutions regarding wave propagation in rectangular waveguide for TM mode. | 10 |
| | (b) | An air filled hollow rectangular waveguide has cross sections of $22.86 \times 10.16 \text{ cm}^2$. Calculate cutoff frequency for dominant mode. | 2 |
| OR | | | |
| Q-5 | (a) | If $Z_L = 100 + j100 \Omega$ and $Z_0 = 25 \Omega$, find the following quantities for the transmission-line circuit using smith chart.
(1) The reflection coefficient at the load.
(2) The input impedance of the line if length is 0.1λ
(3) The distance from the load to the first voltage minimum.
(4) The distance from the load to the first voltage maximum.
(5) SWR | 10 |
| | (b) | Design a quarter wave transmission transformer to match a line having impedance of 35Ω , feeding at 1GHz to a load of 50Ω . | 2 |
| Q-6 | (a) | A load impedance of $100 + j100 \Omega$ is required to be matched a 50Ω coaxial line using short circuited shunt stub at 500MHz. Determine the position and length of stub using smith chart. | 6 |
| | (b) | A rectangular air filled waveguide with dimension $2.104 \text{ cm} \times 1.052 \text{ cm}$ is operated at 10GHz with a dominant mode $TE_{1,0}$. Find f_c, V_g, λ_g . | 6 |
| OR | | | |
| Q-6 | (a) | Explain how two Magic tees work as a Circulator. | 6 |
| | (b) | Explain electron trajectories in magnetron. | 6 |